

Ex. Show that $2^n > n^3$ for $n \geq 10$.

Basis $n=10$
 $2^{10} > 10^3$ So, Basis is Proved.

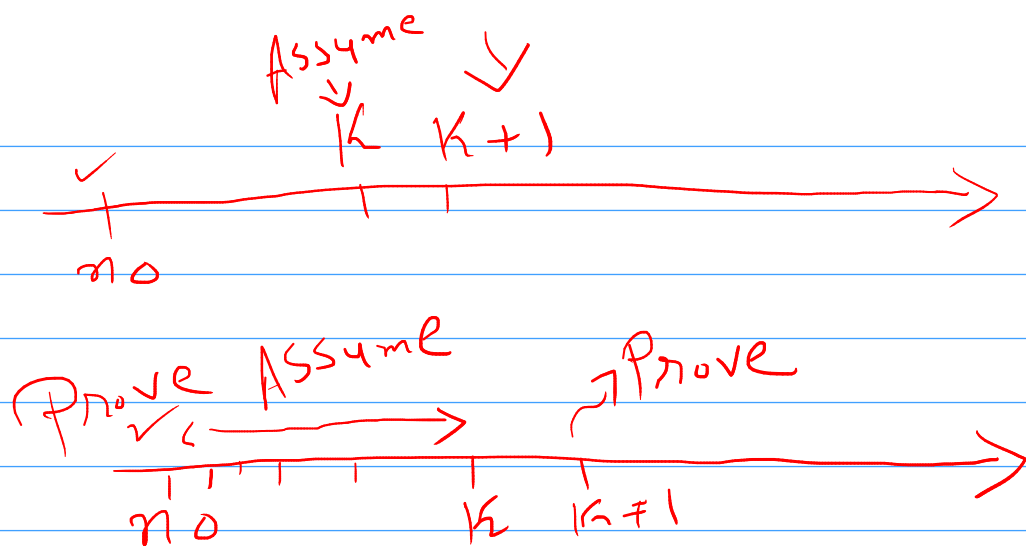
Hypothesis $2^k > k^3 \rightarrow 2^{k+1} > 2k^3$ — (1)

Prove $2^{k+1} > \underline{(k+1)^3}$

R.H.S

$$\begin{aligned}(k+1)^3 &= k^3 + 3k^2 + 3k + 1 \\&\leq k^3 + 3k^2 + 3k^2 + 3k^2 \quad [3k^2 \geq 3k, 3k^2 \geq 1] \\&= k^3 + 9k^2 \\&\leq k^3 + k^3 \quad [k^3 \geq 9k^2] \\&= 2k^3 \\&\leq 2^{k+1} \quad [\text{Result 1}]\end{aligned}$$

$(k+1)^3 < 2^{k+1} \Rightarrow 2^{k+1} > (k+1)^3$



Strong Induction

- Basis: Same as Normal Induction

~ Hypothesis: The statement is assumed to be true for $n_0 \leq n \leq k$.

~ Induction Step: Same as Normal.
 $n = k+1$

~ Strong Induction $\rightarrow$ More Assumption.

[Using Strong Induction]

* Show that any positive integer $n \geq 2$ is either Prime or Product of Prime.

Basis $n=2$
2 is Prime No. So, Stmt is True.

Hypothesis The Statement is True for any n , $2 \leq n \leq k$.
 $k=17$

Induction Step

- Consider a number $(k+1)$

Case-1 $(k+1)$ is Prime, Stmt is True.

Case-2 $(k+1)$ is Not Prime, then $(k+1)$ can be always represented by
 $p \cdot q$ - $[p, q \leq k]$
 $k=23$ $(k+1) = 24$

$$\begin{aligned} 24 &= \overset{p}{8} \times \overset{q}{3} \\ &= 12 \times 2 \\ &= 6 \times 4 \end{aligned} \left\{ \begin{array}{l} p, q \leq k \\ (23) \end{array} \right.$$

- Since $p \leq k$ & $q \leq k$ So, p and q are either Prime or Product of Prime (as per hypothesis)

So, $p \cdot q = (k+1)$ is also Product of Prime or Prime.

* Principle of Inclusion and Exclusion

* Set Cardinality Properties

$$1. |P \cup Q| \leq |P| + |Q|$$

$$2. |P \cap Q| \leq \min(|P|, |Q|)$$

$$3. |P \oplus Q| = |P| + |Q| - 2|P \cap Q|$$

$$4. |P - Q| \geq |P| - |Q|$$

$$|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$$

$$|A_1 \cup A_2 \cup A_3| = |A_1| + |A_2| + |A_3| - |A_1 \cap A_2| - |A_1 \cap A_3| - |A_2 \cap A_3| + |A_1 \cap A_2 \cap A_3|$$

$$|A_1 \cup A_2 \cup \dots \cup A_n| = \sum |Singleton| - \sum |Pairs| + \sum |Triples| - \sum |Quadruples| \dots$$

Principle of Inclusion and Exclusion

$$(-1)^n$$

Ex.
2.12

$$\text{Total Stu} = 200$$

$$|P| = 50 \text{ (Students who take DM)}$$

$$|Q| = 140 \text{ (" " Econ.)}$$

$$|P \cap Q| = 24$$

#

$$\begin{aligned} |P \cup Q| &= |P| + |Q| - |P \cap Q| \\ &= 50 + 140 - 24 \\ &= 166 \end{aligned}$$

$$\begin{aligned} \text{Remaining Students} &= 200 - 166 \\ &= 34 \end{aligned}$$

Ex.

$$\text{Total Cars} = 30$$

$$|A1| = 15 \text{ (Radio)}$$

$$|A2| = 8 \text{ (Ac)}$$

$$|A3| = 6 \text{ (White Wall Tires)}$$

$$\# \quad |A1 \cap A2 \cap A3| = 3$$

~ At least how many Cars do not have any option at all.

$$|A \cup A_2 \cup A_3| = 15 + 8 + 6 - |A \cap A_2| - |A \cap A_3| - |A_2 \cap A_3| + 3$$

$$= 32 - \underbrace{|A \cap A_2|}_{\geq 3} - \underbrace{|A \cap A_3|}_{\geq 3} - \underbrace{|A_2 \cap A_3|}_{\geq 3}$$

$$|A \cap A_2| \geq |A \cap A_2 \cap A_3| \quad (3)$$

$$|A \cap A_2| \geq 3, \quad |A \cap A_3| \geq 3, \quad |A_2 \cap A_3| \geq 3$$

$$|A \cup A_2 \cup A_3| \leq 32 - 3 - 3 - 3$$

$$\leq 23$$

- at most 23 cars have one or more options.

$$\text{Total Cars} = 30$$

$$\text{So, } 30 - 23 = 7 \text{ do not have any option.}$$